

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-25 (AN)** **Date: 24.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

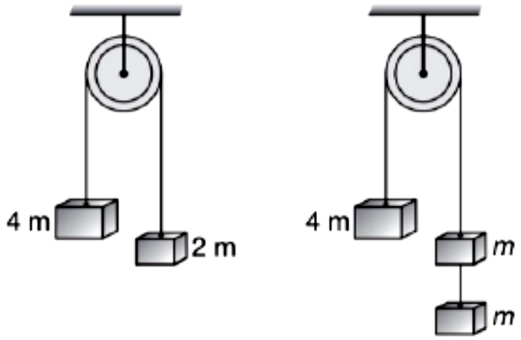
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In the arrangement shown in figure, pulley is smooth and massless and all the strings are light. Let F_1 be the force exerted on the pulley in CASE-1 and F_2 be the force in CASE-2. Then

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A) $F_1 > F_2$ B) $F_1 < F_2$ C) $F_1 = F_2$ D) $F_1 = 2F_2$

2. A spherical ball falls through viscous medium with terminal velocity v. If this ball is replaced by another ball of the same mass but half the radius, then the terminal velocity will be (neglect the effect of buoyancy)

A) v B) $2v$ C) $4v$ D) $8v$

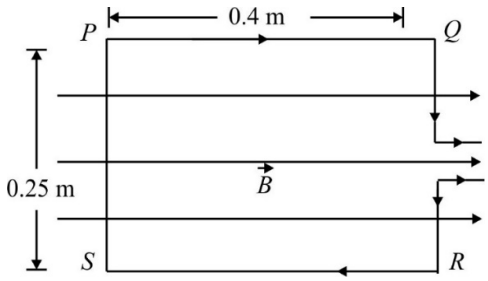
3. Two identical metallic spheres A and B carry charges $+Q$ and $-2Q$ respectively. The force between them is F newton, when they are separated by a distance d in air. The spheres are allowed to touch each other and moved back to their initial positions. The force between them now becomes

A) $\frac{F}{2}$ B) $\frac{F}{4}$ C) $\frac{F}{8}$ D) zero

4. The ratio of the velocities of sound in hydrogen gas and oxygen gas at S.T.P is

A) 16:1 B) 8:1 C) 4:1 D) 2:1

5. A rectangular current carrying coil is placed in a uniform magnetic field $B = 20\,mT$ such that the sides PQ and RS are parallel to B. If the current in the coil is 2A, then the torque on the coil is

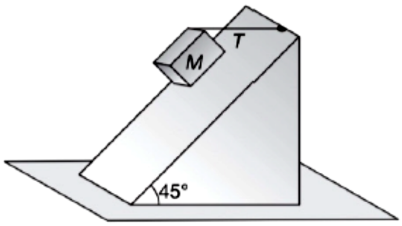


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A) $2\times10^{-3}\,Nm$ B) $4\times10^{-2}\,Nm$ C) $4\times10^{-3}\,Nm$ D) $5\times10^{-3}\,Nm$

6. A block of mass 15 kg is resting on a rough inclined plane as shown in figure. The block is tied up by a horizontal string which has a tension of 50 N. The minimum coefficient of friction between the surfaces of contact is $\left(g=10\,ms^{-2}\right)$



A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{3}{4}$ D) $\frac{1}{4}$

7. 9 identical light bulbs, each designed to extract power P from certain voltage supply are connected in series across the supply. The total power drawn by them is

A) $9P$ B) P C) $\frac{P}{9}$ D) $\frac{P}{81}$

8. The radius R of a nucleus changes with the nucleon number A of nucleus as

A) $R\propto A^{\frac{2}{3}}$ B) $R\propto A^{\frac{1}{3}}$ C) $R\propto A^{0}$ D) $R\propto A$

9. A gas mixture consists of 2 moles of oxygen and 4 moles of argon at temperature T. Neglecting all vibrational modes, the total internal energy of the system is

A) 4 RT B) 15 RT C) 9 RT D) 11 RT

10. The escape velocity for a body projected vertically upwards from the surface of earth is 11 km/s. if the body is projected at an angle of 45^0 with the vertical, the escape velocity will be

A) $11\sqrt{2}\,km/s$ B2) $22\,km/s$ C) $11km/s$ D) $22\sqrt{2}\,km/s$

11. An electromagnetic wave going through vacuum is described by $E=E_0\sin\left(kx-\omega t\right)$; $B=B_0\sin\left(kx-\omega t\right)$. Then

A) $E_0k=B_0\omega$ B) $E_0B_0=\omega k$ C) $E_0\omega=B_0k$ D) $E_0\omega k=B_0$

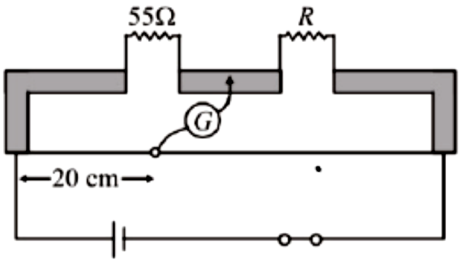
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12. A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ. If μ represents the refractive index of glass with respect to air, then the angle between reflected and refracted rays is

A) $90^\circ + \phi$ B) $\sin^{-1}(\mu\cos\phi)$ C) 90° D) $90^\circ - \sin^{-1}\left(\frac{\sin\phi}{\mu}\right)$

13. Shown in the figure below is a meter-bridge set up with null deflection in the galvanometer. The value of the unknown resistor R is



A) 220 Ω B) 110 Ω C) 55 Ω D) 13.75 Ω

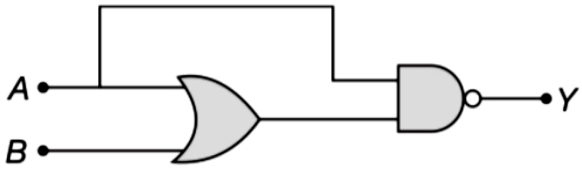
14. Time period of a simple pendulum is T. If its length increased by 2%, the new time period becomes

A) 0.98 T B) 1.02 T C) 0.99 T D) 1.01 T

15. The velocity v of a particle at time t is given by $v = \frac{a}{t} + \frac{bt}{t^2 + c}$. The dimensions of a, b, c are respectively

A) LT^{-2}, L, T B) L, L, T^2 C) L, LT, T^{-2} D) L, L, LT^2

16. The truth table for the circuit given in the figure is



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A)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

C)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

D)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

17. Voltage applied to an AC circuit and the current flowing in it are respectively given by $V = 200\sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)$ and $i = -\sqrt{2}\cos\left(\omega t + \frac{\pi}{4}\right)$

The average power consumed in the circuit is

A) 200 W B) 400 W C) $200\sqrt{2}$ W D) Zero

18. Stopping potential for photoelectrons

A) does not depend on the frequency of the incident light

B) does not depend on the nature of cathode material

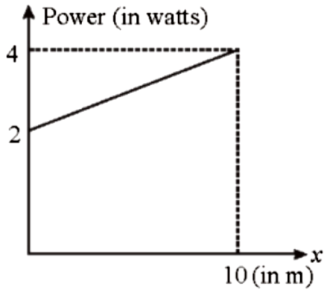
C) depends on both the frequency of the incident light and the nature of the cathode material.

D) depends on the intensity of the incident light.

19. A particle A of mass 10/7 kg is moving in the positive direction of x. Its initial position is $x = 0$ and initial velocity is 1 m/s. The velocity at $x = 10m$ is: (use the graph given)

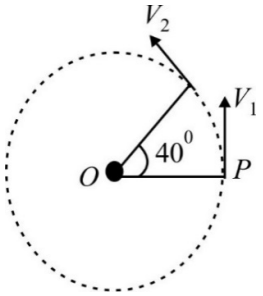
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A) 4 m/s B) 2 m/s C) $3\sqrt{2}\,m/s$ D) $100/3\ m/s$

20. A particle is moving in a circle of radius r centred at O with constant speed v. The change in velocity in moving from P to $Q\left(\angle POQ = 40^{\circ}\right)$ is



A) $2v\cos\left(40^{\circ}\right)$ B) $2v\sin\left(40^{\circ}\right)$ C) $2v\cos\left(20^{\circ}\right)$ D) $2v\sin\left(20^{\circ}\right)$

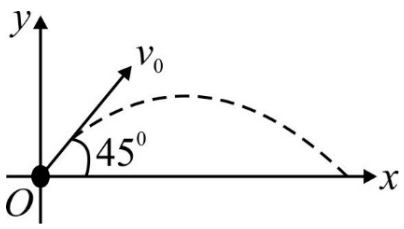
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

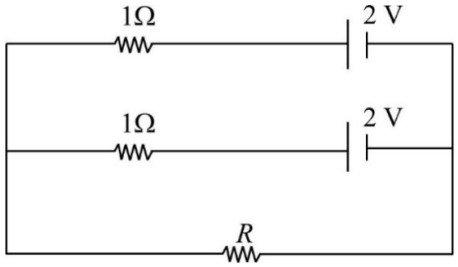
21. A particle is projected at time $t = 0$ from a point O with a speed v_0 at an angle of 45^0 to the horizontal. If the magnitude of the angular momentum of the particle about the point O at time $t = v_0/g$ is $\dfrac{mv_0^{\,3}}{x\sqrt{yg}}$, then the value of $x + y$ is____

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

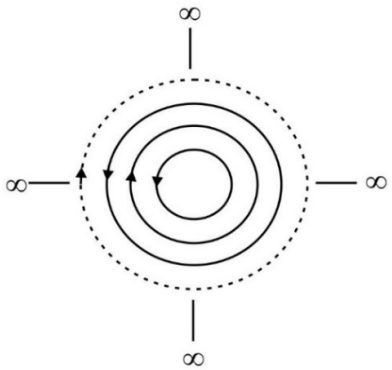


22. Two identical batteries, each of EMF 2 V and internal resistance $r = 1\Omega$ are connected as shown. Find the maximum power that can be developed across R using these batteries (in W)



23. A beam of light strikes a glass sphere of diameter 15 cm which was supposed to converge towards a point 30 cm behind the pole of the spherical surface. Find the position of the image (in cm) if the refractive index of glass is 1.5

24. Infinite conducting rings each having current I in the direction shown are placed concentrically in the same plane as shown in the figure. The radius of rings are $r, 2r, 2^2 r, 2^3 r, ...\infty$. The magnetic field at the centre of rings will be $\frac{\mu_0 I}{nr}$. The value of n is_____

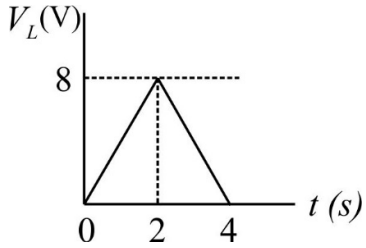


25. The potential difference across a 4H inductor vary with time as shown in Fig. The

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current is zero at time t=0. Find the current (in ampere) at time t=2s.



CHEMISTRY **MAX.MARKS: 100**

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26. Which of the following is correctly matched?

	Column-I	Column-II	Column-III
A)	$[Cr(CO)_6]$	Paramagnetic	Octahedral, sp^3d^2
B)	$[Fe(CO)_5]$	Paramagnetic	Trigonal bipyramidal, sp^3d
C)	$[Co(CO)_4]^+$	Diamagnetic	Square planar, dsp^2
D)	$[Ni(CO)_4]$	Diamagnetic	Square planar, dsp^2

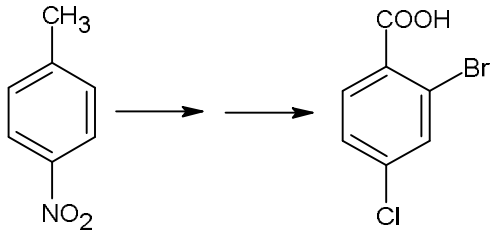
27. Find correct set of quantum numbers of n,l,m,s for last electron of element (atomic number 89).

A) $4,3,0,+\frac{1}{2}$ B) $5,3,0,+\frac{1}{2}$ C) $6,2,0,+\frac{1}{2}$ D) $7,0,0,+\frac{1}{2}$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

28.



In the above conversion the correct sequence of reagents to be added is

A) $(i)Br_2 / Fe, (ii)Fe / H^+, (iii)NaNO_2 / HCl(0-5)^0 C, (iv)CuCl / HCl, (v)KMnO_4$

B) $(i)Br_2 / Fe, (ii)Fe / H^+, (iii)KMnO_4, (iv)Cl_2$

C) $(i)KMnO_4, (ii)Br_2 / Fe, (iii)Fe / H^+, (iv)Cl_2$

D) $(i)Fe / H^+, (ii)HONO, (iii)\ CuCl, (iv)KMnO_4, (v)Br_2$

29. **Statement-I:** I^- and IO_3^- undergo redox reaction to give I_2 and H_2O under acidic condition.

Statement-II: $HCHO$ undergo disproportionation to give $HCOO^-$ and CH_3OH in presence of conc. $NaOH$

A) Both Statement I and Statement II are correct

B) Statement-I is incorrect but Statement II is correct

C) Statement I is correct but Statement II is incorrect

D) Both Statement I and Statement II are incorrect

30. How many of the following are incorrect orders of property indicated against it?

(i) $I^- > Br^- > Cl^- > F^-$ (order of polarizability)

(ii) $Li^+ > Na^+ > K^+ > Cs^+$ (order of polarising power)

(iii) $Li^+ > Mg^{2+} > Al^{3+}$ (order of polarising power)

(iv) LiI > NaI > KI (order of ionic character)

(v) AgI > AgBr > AgCl (order of solubility in water)

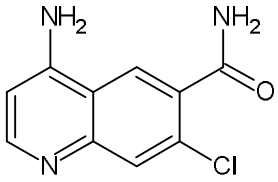
(vi) (Si–Si bond) Si > SiO_2 (Si–O bond) (order of % covalent character of bond.

A) i, ii, vi only B) iii, vi, v only C) iii, iv, v only D) i, ii, iv, v only

APEX JEE-MAIN 2025

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31. An organic compound X shown below is subjected to the Kjeldahl method for the determination of nitrogen. The amount of ammonia liberated from 2.21g of X is?



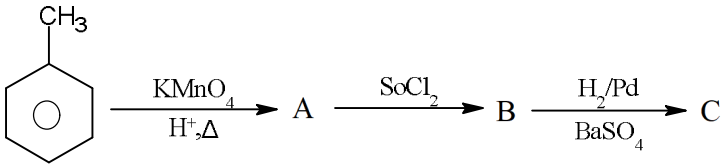
A) 0.34g B) 0.17g C) 0.51g D) 0.68g

32. Amongst $LiCl, BeCl_2, MgCl_2$ and $RbCl$ the compounds with greatest and least ionic character, respectively are:

A) $LiCl$ and $RbCl$ B) $RbCl$ and $BeCl_2$

C) $RbCl$ and $MgCl_2$ D) $MgCl_2$ and $BeCl_2$

33. Identify the name reaction for B to C



A) Gattermann – Koch Reaction B) Rosenmund Reduction

C) Stephen Reaction D) Etard Reaction

34. Which of the following statements are correct?

a) Glycerol is purified by vacuum distillation because it decomposes at its normal boiling point.

b) Aniline can be purified by steam distillation, as aniline is miscible in water.

c) Fractional distillation is used to separate different fractions of crude oil in petroleum industry.

d) compounds with sharp melting and boiling points are considered to be pure.

Choose the most appropriate answer from the options given below.

A) a, c, d only B) b, c, d only C) a, b, c only D) a, b, d only

APEX JEE-MAIN 2025

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35. **Statement-I:** The NH_2 group in aniline is *ortho* and *para* directing and a powerful activating group with respect to electrophilic aromatic substitution reactions.

Statement-II: Aniline does not undergo Friedel-Craft's reaction (alkylation and acylation)

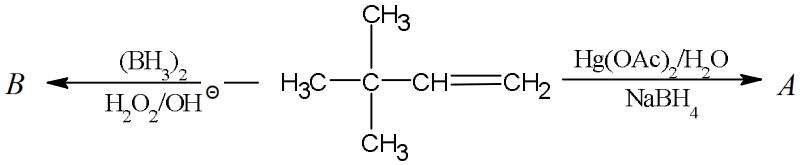
A) Statement-I is true but Statement II is false

B) Both Statement I and Statement II are true

C) Both Statement I and Statement II are false

D) Statement-I is false but Statement II is true

36. Choose the correct option for the following reactions.



A) 'A' and 'B' are both Markownikoff addition products.

B) 'A' is Markownikoff product and 'B' is anti-Markownikoff product.

C) 'A' and 'B' are both anti-Markownikoff addition products.

D) 'B' is Markownikoff product and 'A' is anti-Markownikoff product

37. Consider the reaction, $3MnO_4^{2-} + 4H^+ \longrightarrow$

Then the correct statement(s) regarding this reaction is/are

a) This is a disproportionation reaction.

b) MnO_2 is one of the product of this reaction

c) MnO_4^- is one of the product of this reaction

d) Mn_2O_3 is one of the product of this reaction

A) a, b and d only

B) a, b and c only

C) b and c only

D) a and c only

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

38. Consider the following table:

Compound	**Enthalpy of Formation (magnitude)** $\left(kJ\,mol^{-1}\right)$	**Bond angle** $H\hat{X}H$	**Boiling point** $\left({}^{0}C\right)$
H_2O	x_1	y_1	z_1
H_2S	x_2	y_2	z_2
H_2Se	x_3	y_3	z_3
H_2Te	x_4	y_4	z_4

According to given information select the correct orders from the following:

a) $x_1 > x_2 > x_3 > x_4$ b) $y_1 > y_4 > y_3 > y_2$

c) $z_1 > z_3 > z_2 > z_4$ d) $y_1 > y_2 > y_3 > y_4$

e) $z_1 > z_4 > z_3 > z_2$ f) $x_1 > x_4 > x_3 > x_2$

A) a, b, c B) b, c, d C) b, d, e D) a, d, e

39. At constant temperature, the equilibrium constant $\left(K_p\right)$ for the decomposition reaction $N_2O_4 \rightleftharpoons 2NO_2$ is expressed by $K_p = \left(4x^2P\right)/\left(1-x^2\right)$, where P = total pressure at equilibrium, x = extent of decomposition. Which one of the following statements is true?

A) K_p increases with increase of P

B) K_p increases with increase of x

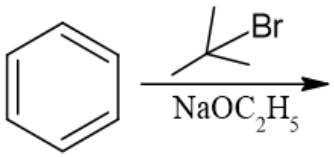
C) K_p increases with decrease of x

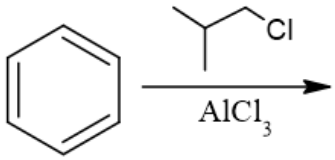
D) K_p remains constant with change in P and x

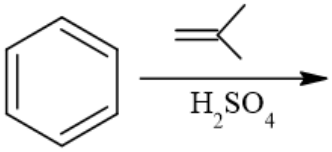
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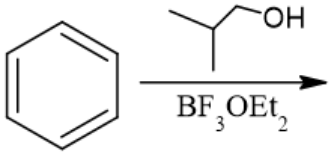
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40. Among the following, reaction which does not give tert-butyl benzene as the major product is

A) 

B) 

C) 

D) 

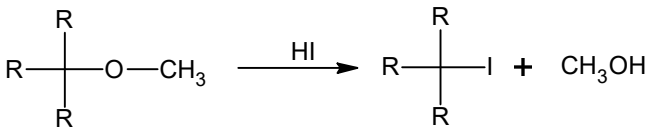
41. Dissolving 1.24 g of white phosphorus in boiling $NaOH$ solution in an inert atmosphere gives a gas PH_3. The amount of $CuSO_4$ (in g) required to completely consume the obtained PH_3 gas as per the following reaction is____

$3CuSO_4 + 2PH_3 \rightarrow Cu_3P_2 + 3SO_3 + 3H_2O$

[Given, atomic mass of H =1, O=16, Na = 23, P=31, S=32, Cu=63]

A) 0.38 B) 2.38 C) 1.38 D) 3.38

42. The cleavage of ether shown below is fastest when



A) One alkyl group is replaced by a phenyl group and other by methyl groups

B) All alkyl groups are replaced by methyl groups

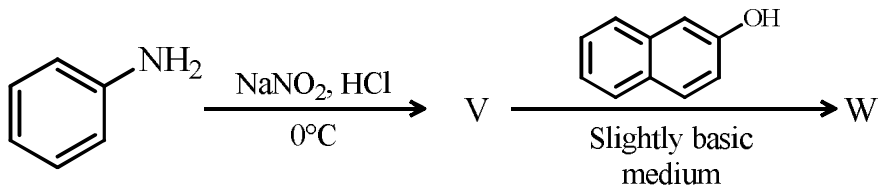
C) One alkyl group is replaced by a p-methoxy phenyl group and others by ethyl groups

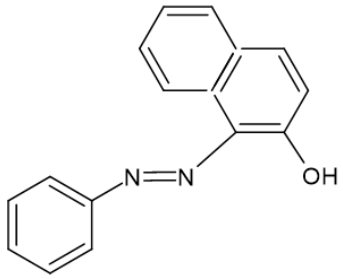
D) One alkyl group is replaced by a p-nitro phenyl group and others by hydrogens

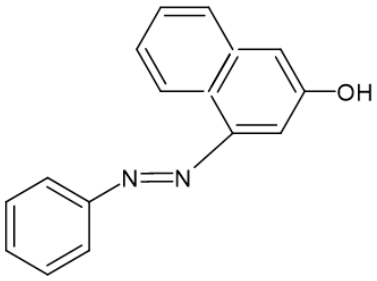
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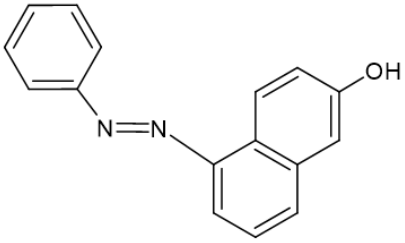
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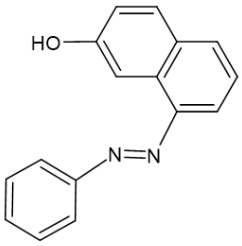
43. In the following reactions, the major product *W* is



A)


B)


C)


D)


44. Match the electronic configurations in List-I with appropriate metal complex ions in List-II and choose the correct option.

[Atomic number: Fe = 26, Mn = 25, Co = 27]

	List-I		**List-II**
(P)	$t_{2g}^6 e_g^0$	(i)	$\left[Fe(H_2O)_6\right]^{2+}$
(Q)	$t_{2g}^3 e_g^2$	(ii)	$\left[Mn(H_2O)_6\right]^{2+}$
(R)	$e_g^2 t_{2g}^3$	(iii)	$\left[Co(NH_3)_6\right]^{3+}$
(S)	$t_{2g}^4 e_g^2$	(iv)	$\left[FeCl_4\right]^{-}$
		(v)	$\left[CoCl_4\right]^{2-}$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

A) $P \to i; Q \to iv; R \to ii; S \to iii$ B) $P \to i; Q \to ii; R \to iv; S \to v$

C) $P \to iii; Q \to ii; R \to v; S \to i$ D) $P \to iii; Q \to ii; R \to iv; S \to i$

45. Which of the following statement(s) is/are correct:

a) phosphorus is more abundant in earth's crust than nitrogen

b) Nitrogen has unique ability to form $p\pi - p\pi$ bonding while phosphorus prefers to form $d\pi - p\pi$ bonds

c) White P is soluble in water and CS_2 while red P is insoluble in both the solvents

d) Red, black phosphorus do not catch fire in air while white phosphorus readily catches fire in air

A) a, b, c only B) b, c, d only C) a, b, d only D) a, b only

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. An aqueous KCl solution of density $1.20\,g\,mL^{-1}$ has a molality of $3.30\,mol\,kg^{-1}$. The molarity of the solution in $mol\,L^{-1}$ is (nearest integer) _____ $\times 10^{-1}$.

[Molar mass of $KCl = 74.5$]

47. An unknown compound (A) (molar mass = 180) on acetylation gives a product (molar mass = 390). Then the number of hydroxyl groups present in one molecule of "A" are____

48. An organic compound undergoes first-order decomposition. The time taken for its decomposition to 1/8 and 1/10 of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\left[\frac{t_{1/8}}{t_{1/10}}\right] \times 10$? $\left(\log_{10} 2 = 0.3\right)$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

49. Consider the expansion of one mole of an ideal gas from an initial state to a final state, with ΔS representing the entropy change in the process. The number of incorrect statement(s) is?

a) If the process is adiabatic, the value of ΔS will be higher in comparision to its irreversible counterpart.

b) If the process is carried out adiabatically, the value of ΔS is zero, irrespective of whether the process is carried out reversibly or irreversibly.

c) If the process is isothermal, there will not be any heat exchange with the surrounding since the temperature of the system and surrounding are equal.

d) If the process is isothermal and the final volume is double of the initial volume, the value of ΔS will be independent of the temperature.

e) If the process is isothermal, the value of ΔS will be higher in irreversible compression than in irreversible expansion

50. In order to oxidise a mixture of two moles each of FeC_2O_4, $Fe_2(C_2O_4)_3$ and $FeSO_4$ in acidic medium, the number of moles of $KMnO_4$ required is_____

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The number of ways in which 17 identical toys can be distributed among 4 children such that each child gets atleast 3 toys, is

A) 120 B) 56 C) 210 D) 104

52. Let $A(\alpha, \beta)$, $B(1, 3)$ and $C(-8, -12)$ respectively denote the centroid, circumcentre and orthocentre of a triangle. The distance of a point $P(\alpha+5,\ 2\beta+8)$ from the line $3x+4y-10=0$ is

A) $\frac{3\sqrt{3}}{5}$ B) $\frac{3}{5}$ C) 3 D) 5

53. Let Z_1 and Z_2 be two complex number such that $Z_1 - Z_2 = 4$ and $Z_1^2 + Z_2^2 = 4 - 6i$. If the value of $Z_1^4 + Z_2^4 = a + ib$ (where $a, b \in R$), then the value of $(a-b)$ is

A) 46 B) 48 C) 50 D) 52

54. If a chord of the circle $x^2 + y^2 - 4x - 2y - c = 0$ is trisected at the points $\left(\frac{1}{3}, \frac{1}{3}\right)$ and $\left(\frac{8}{3}, \frac{8}{3}\right)$, then the radius of the circle will be:

A) 3 B) 4 C) 5 D) 6

55. Let $f : R^+ \to R$ be a differentiable function where

$f(x) = e - (x-1)(\ln x - 1) + \int_1^x f(t)\,dt$, then $\left|f(e) - e^e\right|$ equals

A) 0 B) 2 C) 1 D) 4

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

56. Let $\vec{a},\vec{b},\vec{c}$ be three non-zero vectors satisfying $\vec{a}=\vec{b}\times\vec{c}+2\vec{b}$ where $|\vec{b}|=|\vec{c}|=2$ and $|\vec{a}|\le 4$. The sum of possible value (s) of $|2\vec{a}+\vec{b}+\vec{c}|$ is:

A) 8 B)12 C)20 D)32

57. An ellipse has eccentricity $\frac{1}{2}$ and one focus at the point $P\left(\frac{1}{2},1\right)$. Its one directrix is the common tangent (nearer to the point P) to $x^2+y^2=1$ and the hyperbola $x^2-y^2=1$. The equation of the ellipse is

A) $\frac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{9}}+\frac{(y-1)^2}{\frac{1}{12}}=1$ B) $\frac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{8}}+\frac{(y-1)^2}{\frac{1}{12}}=1$

C) $\frac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{9}}+\frac{(y-1)^2}{\frac{1}{8}}=1$ D) $\frac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{12}}+\frac{(y-1)^2}{\frac{1}{8}}=1$

58. The least non-negative integral value of λ for which the equation $2x^2-2(2\lambda+1)x+\lambda(\lambda+1)=0$ has one root less than λ and other root greater than λ is equal to:

A) 0 B) 1 C) 2 D) 4

59. If $f(x)=\lim\limits_{n\to\infty}\frac{[x]+[2^3.x]+[3^3.x]+.....+[n^3.x]}{n^4}$ $\left(\text{where}[.]\text{ denotes GIF}\right)$, then $f(f(32))$ is

A) 2 B) 4 C) 8 D) 16

60. If $\int_0^x f(t)dt=e^x-ae^{2x}\int_0^1 f(t)e^{-t}dt$, then $f(1)+2f(2)$ is equal to:

A) $e-4e^4$ B) $e-2e^4$ C) $e-2e^2$ D) $2e^2-e^4$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

61. If the mean and standard deviation of the data 3, 5, 7, 'a' and 'b' are 5 and 2 respectively, then a and b are the roots of the equation

A) $x^2-20x+18=0$ B) $2x^2-20x+19=0$

C) $x^2-10x+18=0$ D) $x^2-10x+19=0$

62. Consider the function $f:R\to R$, $f(x)=\sqrt[3]{x+\sqrt{1+x^2}}+\sqrt[3]{x-\sqrt{1+x^2}}$. If $\sum_{r=1}^{n} f^{-1}(r)=\frac{n^2(n+1)^2}{\lambda}+\frac{3n(n+1)}{\mu}$, then value of $(\lambda+\mu)$ is

A) 4 B) 6 C) 8 D) 12

63. If $\int x^{26}(x-1)^{17}(5x-3)dx=\frac{x^{27}(x-1)^{18}}{k}+C$ (where C is constant of integration), then the value of k is:

A) 3 B) 6 C) 9 D)12

64. If $\lim_{x\to 0}\frac{10-\sum_{k=1}^{10}(\cos kx)}{x^2}=\frac{a}{b}$ (where a and b are co-prime), then the value of $(a+b)$ is equal to:

A) 38 B) 385 C) 386 D) 387

65. Let $S_n=\sum_{r=1}^{n} r!$ and for $n>6$, $x=S_n-7\left[\frac{S_n}{7}\right]$ (where $[.]$ denotes GIF). If $\sin^{-1}(\sin x)=a+b\pi$ (where $a,b\in Z$), then $|a^2-b^2|$ is

A) 20 B) 21 C) 22 D) 23

66. A biased coin has a property that probability of occurrence of head is twice that of tail. Probability that if the coin is tossed twice then the head occurs atleast once is_____

A) $\frac{8}{9}$ B) $\frac{1}{6}$ C) $\frac{1}{3}$ D) $\frac{4}{9}$

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

67. Let $A=\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ 3 & 2 & 1\end{bmatrix}$ and U_1, U_2 *and* U_3 are columns of a 3×3 matrix U. If column matrices U_1, U_2 *and* U_3 satisfies $AU_1=\begin{bmatrix}1\\0\\0\end{bmatrix}$, $AU_2=\begin{bmatrix}2\\3\\0\end{bmatrix}$, $AU_3=\begin{bmatrix}2\\3\\1\end{bmatrix}$, then the value of $\det(U)$ is

A) 0 B) 1 C) 3 D) 5

68. $\left|\frac{8}{\pi^2}\int_0^{\pi}\frac{x\sin^4 x}{\sin^4 x+\cos^4 x}dx\right|$ is equal to

A) 4 B) 8 C) 32 D) 2

69. A line 'L' is drawn from (4, 3) to meet the line $L_1: 3x+4y+5=0$ and $L_2: 3x+4y+15=0$ at point A and B respectively. From 'A', a line perpendicular to L is drawn meeting the line L_2 at A_1. Similarly from point 'B', a line perpendicular to L is drawn meeting the line L_1 at B_1. Thus a parallelogram AA_1BB_1 is formed. The slope(s) of 'L' so that the area of parallelogram AA_1BB_1 is least are m_1 and m_2 (where $m_1<0$ and $m_2>0$). The value of $\left|\frac{m_1}{m_2}\right|$ is ____

A) $\frac{1}{49}$ B) 49 C) 1 D) $\frac{1}{7}$

70. In the expansion of $(1+x)^m(1-x)^n$, the coefficients of x and x^2 are 3 and –6 respectively. The value of n is_____ $(m,n\in N)$

A) 9 B) 3 C) 6 D) 7

APEX JEE-MAIN 2025

For More Material Join: @JEEAdvanced_2025

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. Consider a matrix $A=\begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{bmatrix}$, where α, β, γ are three distinct natural numbers. If $\frac{\det\left(adj\left(adj\left(adj\left(adj\ A\right)\right)\right)\right)}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}}=2^{32}\times 3^{16}$, then the number of such ordered triplets (α, β, γ) is_____

72. Let $\overline{a}=-\hat{j}-\hat{k}$, $\overline{b}=-\hat{i}+\hat{j}$ and $\overline{c}=\hat{i}-2\hat{j}+3\hat{k}$. If $\overline{r}$ is any vector such that $\overline{r}\times\overline{b}=\overline{c}\times\overline{b}$ and $\overline{r}.\overline{a}=0$, then the value of $\left|\overline{r}.\overline{b}\right|$ is

73. Consider $f(x)=\frac{\sin x+ae^{x}+be^{-x}+c\ln(1+x)}{x^3}$, where a, b, c are real numbers. If $\lim\limits_{x\to 0^+} f(x)$ is finite, then the value of a + b – c is

74. Let $\frac{dy}{dx}=\frac{y\phi'(x)-y^2}{\phi(x)}$, where $\phi(x)$ is a function satisfying $\phi(1)=1$, $\phi(4)=1296$. If $y(1)=1$, then $y(4)$ is equal to _____

75. Let $R=\{(1,2),(2,3)\}$ be a relation defined on set {1, 2, 3}. The minimum number of ordered pairs required to be added in R such that enlarged relation becomes an equivalence relation is___

APEX JEE-MAIN 2025

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